Chi Square Based Measures

Lecture 8

FOLLOWING-UP CHI-SQUARED TESTS

- Like any significance test, chi-squared tests of independence have limited usefulness.
- A small *P-value indicates strong evidence of association but* provides little information about the nature or strength of the association.
- Statisticians have long warned about dangers of relying solely on results of chi-squared tests rather than studying the nature of the association
- we discuss ways to follow up the tests to learn more about the association.

Phi Coefficient

- The phi correlation coefficient (phi) is one of a number of correlation statistics developed to measure the strength of association between two variables.
- The phi is a nonparametric statistic used in cross-tabulated table data where both variables are dichotomous.
- *Dichotomous* means that there are only two possible values for a variable.

Phi Coefficient

- Pearson Chi-Square provides information about the existence of relationship between 2 nominal variables, but not about the magnitude of the relationship.
- Phi coefficient is the measure of the strength of the association

$$\phi = \sqrt{\frac{\chi^2}{N}}$$

OR

$$\boldsymbol{\Phi} = \frac{(\boldsymbol{AD} - \boldsymbol{BC})}{\sqrt{(\boldsymbol{A}+\boldsymbol{B})(\boldsymbol{C}+\boldsymbol{D})(\boldsymbol{A}+\boldsymbol{C})(\boldsymbol{B}+\boldsymbol{D})}}$$

Interpretation

Similar to a Pearson Correlation Coefficient, a Phi Coefficient takes on values between -1 and 1 where:

- **-1** indicates a perfectly negative relationship between the two variables.
- **0** indicates no association between the two variables.
- **1** indicates a perfectly positive relationship between the two variables.

In general, the further away a Phi Coefficient is from zero, the stronger the relationship between the two variables.

Limitations of Phi- Coefficient

- The measure is not very satisfactory in case of 2x2 table, since it does not necessarily have an upper '+1' or lower limit '-1'.
- This can only happen when entries at b=c=0 or a=d=0 then Phi coefficient is -1 or +1.

Example

	Nature of Relation and Sample Size					
	Strong, $n = 150$		Weak, $n = 150$		Weak, $n = 600$	
Opinion	Male	Female	Male	Female	Male	Female
Agree	75	25	65	25	260	100
Disagree	25	25	35	25	140	100
χ^2	9.375		3.125		12.500	
n	150		150		600	
ϕ^2	0.0625		0.02083		0.02083	
ϕ	0.25		0.144		0.144	

Contingency coefficient

A slightly different measure of association is the contingency coefficient. This is another chi square based measure of association, and one that also adjusts for different sample sizes. The contingency coefficient can be defined as

$$C = \sqrt{\frac{\chi^2}{n + \chi^2}}.$$

- The contingency coefficient has much the same advantages and disadvantages as does ϕ. When there is no relationship between two variables, C = 0.

Limitations of Coefficient of contingency

- The contingency coefficient may be less than 1 even when two variables are perfectly related to each other. This means that it is not as desirable a measure of association as those which have the range 0 to 1.

Example

	Nature of Relation and Sample Size					
	Strong, $n = 150$		Weak, $n = 150$		Weak, $n = 600$	
Opinion	Male	Female	Male	Female	Male	Female
Agree	75	25	65	25	260	100
Disagree	25	25	35	25	140	100
χ^2	9.375		3.125		12.500	
n	150		150		600	
C	0.2425		0.1429		0.1429	

Cramer's V

- When the table is larger than 2 by 2, a different index must be used to measure the strength of the relationship between the variables. One such index is Cramer's V.
- If Cramer's V is large, it means that there is a tendency for particular categories of the first variable to be associated with particular categories of the second variable.

$$V = \sqrt{\frac{\chi 2}{nt}}$$

t = Minimum (r − 1, c − 1).

- By using the information concerning the dimensions of the table, Cramer's V corrects for the problem that measures of association for tables of different dimension may be difficult to compare directly.
- Cramer's V equals 0 when there is no relationship between the two variables, and generally has a maximum value of 1, regardless of the dimension of the table or the sample size.

- This makes it possible to use Cramer’s V to compare the strength of association between any two cross classification tables. Tables which have a larger value for Cramer’s V can be considered to have a strong relationship between the variables, with a smaller value for V indicating a weaker relationship.

Example

Attitude G12	Provincial Political Preference Liberal	NDP	Conservative	Total
Strongly Agree	31 (44.9)	227 (168.5)	31 (75.6)	289
Somewhat Agree	31 (25.2)	77 (94.5)	54 (42.4)	162
Somewhat Disagree	18 (11.5)	19 (43.1)	37 (19.3)	74
Strongly Disagree	8 (6.4)	7 (23.9)	26 (10.7)	41
Total	88	330	148	566

- $\chi 2$ = 126.105
- n = 566.
- r = 4 rows
- c = 3 columns
- r−1 = 4−1 = 3
- c−1 = 3−1 = 2.
- In determining V, we need is the smaller of these, so t= c-1= 2.

- Once the χ 2 value has been calculated, the determination of V is relatively straightforward.

$$V = \sqrt{\frac{\chi\ 2}{nt}} = 0.334$$

The relationship is not much strong.

Summary of Chi Square Based Measures

- Each of ϕ 2 , C and V are useful measures of association. When there is no relationship between the two variables, each of these measures has a value of 0. As the extent of the relationship between the variables increases, each of these measures also increases, although by different amounts.
- Where these measures differ is in their maximum value.

- If there is a perfect relationship between the two variables of a cross classification table, it would be preferable to have the measure of association have a value of 1.
- ϕ 2 can have values exceeding 1 in the case of tables with more than 2 rows and 2 columns. In contrast, the contingency coefficient C sometimes has a maximum value which is less than 1.

- Cramer’s V is the preferred measure among these χ 2 based measures. It generally has a maximum value of 1 when there is a very strong relationship between two variables. When it is computed, Cramer’s V takes account of the dimensions of the table, implying that V for tables of different dimensions can be meaningfully compared.